

RESEARCH STATEMENT

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I work in number theory and representation theory, with a focus on automorphic representations, their local and global asymptotic properties, and the cohomology of arithmetic groups. I am especially motivated by problems in which automorphic forms interact with other areas of mathematics, for example:

Topology: In my thesis [GG23] and subsequent works [DGG22, EGGG23], we bound the dimensions of the cohomology of arithmetic manifolds as they vary in sequences of covering spaces known as towers. For example:

Theorem 1 ([GG23]). *Let p be a large enough prime, N be odd, and $X(p^n)$ be a tower of full-level compact arithmetic manifolds associated to $U(N-a, a)$. In the smallest nontrivial degree of cohomology a , we have:*

$$\dim H^a(X(p^n), \mathbf{Q}) \ll \text{Vol}(X(p^n))^{\frac{N}{N^2-1}}.$$

We derive such results from asymptotic bounds on automorphic representations, thereby proving several instances of the 30-year old Conjecture 1 quantifying the failure of the naïve Ramanujan conjecture.

Fourier analysis: In the collaboration [GGV], we give a new proof, using the cohomology of arithmetic groups, of the following Fourier interpolation formula of Radchenko-Viazovska:

Theorem 2 ([RV19]). *Let f be an even Schwartz function on the real line, and $\hat{f}$ its Fourier transform. The function f is entirely determined by the values $(f(\sqrt{n}), \hat{f}(\sqrt{n}))_{n \in \mathbf{N}}$.*

The starting point for this strategy is the observation that the space $\mathcal{S}$ of Schwartz functions carries the Weil representation of the metaplectic group. We deduce Theorem 2 from the vanishing of a cohomology group $H^1(\Gamma, \mathcal{S}^*)$ for a certain Γ , which we prove using the spectral theory of automorphic forms.

Outline. Section 1 discusses the relation between the cohomology of arithmetic groups and automorphic forms, and lays out recent progress and future directions in the program to use Langlands functoriality to understand their asymptotics. Section 2 concerns cohomology of arithmetic groups with infinite-dimensional coefficients, and applications to interpolation theorems. Finally, Section 3 describes two collaborative projects developed through research workshops, and the role I have now taken as a leader in such research groups.

1. GROWTH OF COHOMOLOGY, LIMIT MULTIPLICITY, AND STATISTICS OF NON-TEMPERED REPRESENTATIONS

A classical measure of the topological complexity of a space X are the cohomology groups $H^i(X, \mathbf{C})$. These groups are studied geometrically using simplices or differential forms, but also algebraically as the *group cohomology* $H^i(\pi_1(X), \mathbf{C})$ of the *fundamental group* $\pi_1(X)$ of the space X .

An especially rich class of spaces are *arithmetic manifolds* X_Γ , whose fundamental groups Γ are arithmetic groups, i.e. the integer entries $G(\mathbf{Z})$ of a (reductive) matrix group G such as GL_N . Some key properties are:

- The spaces X_Γ often parameterize objects of interest, such as quadratic forms or abelian varieties.
- The fundamental groups Γ have natural sequences of *congruence subgroups* $\Gamma(n)$, consisting of matrices congruent to the identity modulo n . The spaces $X_{\Gamma(n)}$ are *covering spaces* of X_Γ .
- Natural families of operators T_p act on $H^i(\Gamma, \mathbf{C})$ and their eigenvalues are expected to be connected to Galois representations. As such, the cohomology is the generalization of *modular forms*.

The cohomology of arithmetic manifolds, though well-studied, is in general very hard to compute. A more tractable problem is to consider the question asymptotically.

Question 1. In a tower $X_{\Gamma(n)}$ of arithmetic manifolds, how does $H^i(\Gamma(n), \mathbf{C})$ grow as a function of n ?

Arithmetic manifolds are Riemannian: their cohomology can be computed using harmonic differential forms, which carry $G(\mathbf{R})$ -representations. For X_Γ compact, Matsushima [Mat67] leveraged this to relate cohomology and representations:

$$(1) \quad H^*(\Gamma, \mathbf{C}) = \oplus_\pi m(\pi, \Gamma) H^*(\mathfrak{g}, K; \pi).$$

This equality expresses the cohomology in terms of the multiplicity $m(\pi, \Gamma)$ in $L^2(\Gamma \backslash G)$ of the irreducible *cohomological* unitary representations π of $G(\mathbf{R})$ (by definition, those such that $H^*(\mathfrak{g}, K; \pi) \neq 0$). A given group G has only finitely many cohomological representations [VZ84], allowing us to rephrase Question 1.

Question 2. Given a representation π of $G(\mathbf{R})$, how does $m(\pi, n) := m(\pi, \Gamma(n))$ grow as a function of n ?

First progress towards this question was made by deGeorge-Wallach [dGW78], who compared $m(\pi, n)$ to the volume of the arithmetic manifolds $X_{\Gamma(n)}$:

$$(2) \quad \lim_{n \rightarrow \infty} \frac{m(\pi, n)}{\text{Vol}(X_{\Gamma(n)})} = \begin{cases} k(\pi) > 0 & \text{if } \pi \hookrightarrow L^2(G(\mathbf{R})) \\ 0 & \text{otherwise.} \end{cases}$$

In particular, they show that the multiplicity $m(\pi, n)$ is always $O(\text{Vol}(X_{\Gamma(n)}))$, or, equivalently, $O(n^{\dim G})$. Additionally, integrability properties of π dictate two regimes for the rate of growth. Building on this second fact, Sarnak-Xue [SX91] conjectured growth rates for unitary representations $\pi \not\hookrightarrow L^2(G(\mathbf{R}))$:

Conjecture 1 (Sarnak-Xue). Let $p(\pi) = \inf\{p \geq 2 \mid \pi \hookrightarrow L^p(G(\mathbf{R}))\}$. Then

$$m(\pi, n) \ll_\epsilon \text{Vol}(X_{\Gamma(n)})^{2/p(\pi)+\epsilon}.$$

Representations such that $p(\pi) > 2$ are said to be *non-tempered*. To discuss them, we adopt a global perspective and consider automorphic representations, i.e. irreducible $G(\mathbf{A})$ -subrepresentations of

$$(3) \quad L^2(G(\mathbf{Q}) \backslash G(\mathbf{A})) = \oplus_\pi m(\pi) \pi$$

where $\mathbf{A}$ are the adèles of $\mathbf{Q}$. Automorphic representations factor as tensor products $\pi = \otimes'_v \pi_v$ of representations of the local groups $G(\mathbf{Q}_v)$; to underscore our shift in perspective, representations of the Lie group $G(\mathbf{R})$ are now denoted π_∞ . The global and classical perspectives are related since up to finite index,

$$\Gamma(n) \backslash G(\mathbf{R}) = G(\mathbf{Q}) \backslash G(\mathbf{A}) / K_f(n)$$

for suitable subgroups $K_f(n) \subset G(\mathbf{A})$. The multiplicity counts of Question 2 can then be rewritten by considering the finite part $\pi_f := \otimes'_{v < \infty} \pi_v$ of the representations, and studying their fixed vectors $\pi_f^{K_f(n)}$. For π_∞^0 a fixed $G(\mathbf{R})$ -representation, the multiplicity is then $m(\pi_\infty^0, n) := \sum_{\pi = \pi_\infty^0 \otimes \pi_f} m(\pi) \dim \pi_f^{K_f(n)}$.

Arithmetically, the richest part of the decomposition (3) and of generalizations to non-compact quotients is the *cuspidal* part, which is the subject of a central conjecture in the analytic theory of automorphic forms:

Conjecture 2 (Ramanujan-Petersson). If G is split and π is cuspidal, the constituents π_v of π are tempered.

This conjecture has analytic implications in number theory and beyond [Li20]. Yet for general G , it fails: the first counterexamples appeared in [Kur78]. But not all is lost: Conjecture 1 can be rephrased as

The Ramanujan-Petersson Conjecture holds asymptotically as $n \rightarrow \infty$.

1.1. Results. My research program aims to prove Conjecture 1 for cohomological representations using Langlands functoriality. The main idea is:

Multiplicities of non-tempered representations grow slowly because these representations come from smaller groups.

To be more precise, we rapidly outline the web of conjecture known as the *Langlands Program*. To any group G , it attaches a dual group $\hat{G}$. The *Reciprocity Conjecture* then predicts a correspondence

$$(4) \quad \{\text{Automorphic representations } \pi_\rho \text{ of } G\} \leftrightarrow \{\text{Representations } \rho_\pi \text{ of } \text{Gal}(\bar{\mathbf{Q}}/\mathbf{Q}) \text{ into } \hat{G}\}.$$

If a Galois representation ρ_π factors through a subgroup $\hat{H} \subset \hat{G}$, then (4) predicts a corresponding π_ρ^H for H , and according to the Langlands *Functoriality Conjecture* the characters of π_ρ^G and π_ρ^H should be related.

A well-understood instance of these relations is the Endoscopic Classification of Representations (ECR), proved by Arthur [Art13] for SO_N and Sp_N , and extended to unitary groups in [Mok15] and [KMSW14]. It describes the decomposition (3) in terms of the contributions of smaller, *endoscopic* groups.

In the footsteps of [MS19] who use endoscopy to prove Conjecture 1 for $U(N-1, 1)$, I prove in my thesis:

Theorem 3 ([GG19],[GG23]). *Let $U(N-a, a)$ be a unitary group of odd rank defined with respect to a CM extension E/F of number fields, let $\mathfrak{p}$ be a prime of F , and let $\Gamma(\mathfrak{p}^n)$ be a sequence of cocompact congruence subgroups. Then for the smallest degree a whose cohomology does not identically vanish,*

$$\dim H^a(\Gamma(\mathfrak{p}^n), \mathbf{C}) \ll |\mathfrak{p}^n|^N,$$

where $|\cdot| : F \rightarrow \mathbf{Q}$ is the norm. In particular, Conjecture 1 holds for the corresponding representations.

The idea of the theorem is to not only appeal to the ECR, but also repurpose its tools, namely the *stabilization of the trace formula*. This is an expression for the multiplicity $m(\pi_\infty, \mathfrak{p}^n)$ in terms of a *stable* contribution of our main group G and of “error terms” coming from the smaller endoscopic groups H .

$$(5) \quad m(\pi_\infty, \mathfrak{p}^n) = St_{\pi_\infty}^G(\mathfrak{p}^n) + \sum_{H \text{ endoscopic}} St_{\pi_H}^H(\mathfrak{p}^n).$$

The main observation powering Theorem 3 is:

For good π_∞ , one of the “error terms” in (5) is in fact the dominant term as n grows.

This, combined with the first observation after (2), allows me to bound the multiplicities in terms of the smaller rate of growth $|\mathfrak{p}^n|^{\dim H}$. In subsequent joint work with Rahul Dalal, we vastly extend the results:

Theorem 4 ([DGG22]). *Let G be a rank N unitary group defined with respect to a CM extension E/F . For any cohomological π_∞ , there is $R(\pi_\infty)$ explicit such that for any sequence of ideals $\mathfrak{n}$ split in E :*

- (i) $m(\pi_\infty, \mathfrak{n}) \ll |\mathfrak{n}|^{R(\pi_\infty)}$ and the exponent $R(\pi_\infty)$ proves Conjecture 1 if $N > 4$.
- (ii) Under some conditions on E and π_∞ , $R(\pi_\infty)$ is sharp. For example, if $G(\mathbf{R}) = U(N-a, a)$ with N odd and a the smallest degree whose cohomology does not identically vanish, then there exist sequences of cocompact lattices $\Gamma(\mathfrak{n})$ such that

$$H^a(\Gamma(\mathfrak{n}), \mathbf{C}) \asymp \text{Vol}(X_{\Gamma(\mathfrak{n})})^{\frac{N}{N^2-1}}.$$

In fact, we prove more refined statistics: we weigh multiplicities by unramified functions and show that the finite factors π_v of representations such that π_∞ and E satisfy the assumptions of (ii) distribute on average according to a Sato-Tate measure predicted by Langlands functoriality.

The proof of Theorem 4 builds on that of Theorem 3, but tackles the much more delicate cases where the G -term in (5) dominates. Inspired by [Taï17], we then analyze this term by taking seriously the intuition:

Endoscopy is the echo for classical groups of parabolic induction happening on GL_N .

In practice, this means that the main work of Theorem 4 is controlling traces of representations obtained by *twisted parabolic induction*. We give upper bounds on these in general, and exact identities in some cases.

What about non-unitary groups? I begin to tackle this question with Shai Evra and Henrik Gustafsson:

Theorem 5 ([EGGG23]). *Let F be a totally real number field, and let $G = SO_5/F$ or an inner form unramified at the finite places.*

- (i) *Conjecture 1 holds for cohomological representations of G . In particular, if $\Gamma(\mathfrak{n})$ is a sequence of cocompact lattices in $G(\mathbf{R})$, then for the unique degree d of interest we have:*

$$\dim H^d(\Gamma(\mathfrak{n}), \mathbf{C}) \ll |\mathfrak{n}|^5, \quad d = \begin{cases} 2 & \text{if } G(\mathbf{R}) = SO(3, 2) \\ 1 & \text{if } G(\mathbf{R}) = SO(4, 1). \end{cases}$$

- (ii) *If the group G is compact at the infinite places, the analogue of Conjecture 1 for finite places v also holds: the multiplicity of non-tempered π_v are negligible as the level grows.*

The generalization of Conjecture 1 in (ii) is the *Sarnak-Xue Density Hypothesis*: its proof opens new applications in which arithmetic manifolds are replaced by discrete analogues such as simplicial complexes. As a corollary of Theorem 5, we exhibit infinite families of so-called Density Ramanujan graphs: regular graphs with optimal mixing properties, which are sought after for applications to computer science [Lub18].

Theorem 5 is also proved in the framework of functoriality, but adopts a mixed approach in which the multiplicity counts are split into a *global* and a *local* components, and both are estimated independently. The estimates rely on Schmidt’s explicit description of the automorphic representations of SO_5 [Sch20].

To go beyond low-rank cases and implement the local/global strategy on general on higher-rank groups requires stronger, more uniform estimates on the asymptotic properties of unitary representations of reductive p -adic groups. This is exactly what Dalal and I prove, in a joint paper with Simon Marshall:

Theorem 6 ([DGGM25]). *Let F_v be a p -adic field and π_v a smooth representation of $GL_N(F)$. Let $K_v(n)$ be the factor of $K_f(n)$ at the place v . Then*

- (1) *The dimension of $K_v(n)$ -fixed vectors in π_v is uniformly bounded in terms of π_v ’s inducing data.*
- (2) *If π_v is additionally unitary, its character Θ_{π_v} admits an uncertainty principle: its decay properties at infinity are quantifiably dictated by its singularities near the origin and vice versa.*

The second result, though not strictly necessary for global counts, can be interpreted as a local version of the Sarnak-Xue conjecture. It points towards yet unexplained symmetries of characters of p -adic representations.

Finally, the framework developed to tackle the limit multiplicity question can be leveraged to answer finer arithmetic questions about automorphic representations. In the last result of this section, we consider the root number of self-dual symplectic¹ cuspidal automorphic representations of GL_{2N} . Our interest in these objects arises from the ECR: among all possible candidate representations of an adelic classical group, the ones that actually arise as an automorphic transfer from GL_N are determined by their root numbers. With a view towards understanding this phenomenon, Dalal and I prove:

Theorem 7 ([DGG24]). *Let F be totally real and let $\mathcal{A}(N, \lambda, n)$ be the set of self-dual symplectic cuspidal automorphic representations of $GL_{2N}(\mathbf{A}_F)$ with infinitesimal character λ and conductor n . If n is large enough, the root numbers of $\pi \in \mathcal{A}(N, \lambda,)$ equidistribute as $\lambda \rightarrow \infty$.*

This recovers the leading term in results of Kimball Martin [Mar23] for modular forms and $GL_2/\mathbb{Q}$. The techniques build on a generalization to higher-rank groups of a classical result:

The root number of a newform is the eigenvalue of the Atkin-Lehner operator.

Building on this intuition, the key inputs into our endoscopic framework are a modification of the functional equation of the Rankin-Selberg L -function, and explicit and uniform description of the space of oldforms by Reeder, a spectral construction of test functions following Rogawski’s twisted Paley-Weiner theorem, and the construction and properties of a stable Plancherel measure.

1.2. Future Work.

Sharp bounds. In upcoming joint work with Dalal and my postdoc Yi Shan [DGGS26], the local results of [DGGM25] will serve as an input to establish sharp upper bounds for counts of automorphic representations, as conjectured in [DGG22]. The key strategy is to uniformly control the dimensions of fixed vectors in so-called *Speh* representation. For lower bounds, the goal is to bypass the subtlety introduced by the root numbers. This is done in a technique inspired by [Mar14]: by fixing an auxiliary level and proving a surjectivity result about local A -packets, we exhibit enough automorphic representations to meet the upper bound. For unweighted limit multiplicity counts, these techniques should extend beyond unitary groups: a key input is a twisted fundamental Lemma for congruence subgroups established by Ganapathy-Varma [GV17].

¹Note that the root numbers of orthogonal self-dual representations are automatically 1.

Applications to counts of cycles. The exact bounds described in the previous paragraph let us apply the framework developed by Bergeron-Millson-Moeglin [BMM16] relating the geometric theta lift and the ECR to deduce results on geometric cycles on compact Shimura varieties. More precisely, we establish an asymptotic variant of the Hodge conjecture when $G = U(n, 1)$ in degrees below the middle, confirming that for compact quotients of the complex ball, asymptotically, 100% of cycles come from cup products of divisors with special cycles.

Explicit transfer and applications. The question of understanding endoscopic transfer explicitly is central to applications, but also an inherently interesting one: the Fundamental lemma is the quintessential example of such an explicit transfer. In joint work with Brooke Feigon, we are working on establishing new examples of explicit transfer of test functions for certain indicator functions of compact-open subgroups, following work of Atobe-Oi-Yasuda [AOY24] and Kazhdan-Varshavsky [KV12]. Some of our motivating applications include depth-preservation results in the ECR, unconditional constructions of new Ramanujan graphs, and establishing connections with the representation theory of finite groups of Lie type.

Uncertainty and non-tempered representations. The second local result of Theorem 6 is deduced combinatorially from the Bernstein-Zelevinsky classification of unitary representations of GL_N , but I expect it to have a harmonic-analytic proof. In this more exploratory research direction, I aim to explain the phenomenon of uncertainty by developing an integral transform on the space of Schwartz functions on the group $GL_N(F)$, for which, as in classical Fourier analysis, there is a tradeoff between pointwise singularities of a distribution (corresponding here to genericity of the representation) and integrability properties at infinity (temperedness). If successful, such a strategy could pave the way to development of a global theory, with possible applications to new approaches towards the Ramanujan conjecture.

2. COHOMOLOGY OF ARITHMETIC GROUPS AND INTERPOLATION

Recently, as part of the series of works leading up to the sphere-packing results for the E_8 and Leech lattices, Radchenko-Viazovska proved the following interpolation theorem.

Theorem 8 ([RV19]). *Let f be an even Schwartz function on the real line, and $\hat{f}$ its Fourier transform. The function f is entirely determined by the values $(f(\sqrt{n}), \hat{f}(\sqrt{n}))_{n \in \mathbf{N}}$, and the only relation comes from the Poisson summation formula.*

2.1. Results. In joint work with Akshay Venkatesh, we give a new proof of this result in the language of the cohomology of arithmetic groups. The main insight driving this proof is:

The space $\mathcal{S}$ of Schwartz functions carries the Weil representation of the metaplectic group.

The metaplectic group is the degree 2 cover of SL_2 , and the Weil representation is defined by:

- $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ multiplies $f(x)$ by $e^{t\pi i x^2}$ and $\begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$ multiplies $\hat{f}(x)$ by $e^{t\pi i x^2}$, and
- the lift of $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ acts by the Fourier transform.

The extension of these formulas to the dual $\mathcal{S}^*$ of the Weil representation, together with a Mayer-Vietoris-type argument, allow us to reduce Theorem 8 to

$$(6) \quad H^0(\Gamma, \mathcal{S}^*) = \mathbf{C}, \quad \text{and} \quad H^1(\Gamma, \mathcal{S}^*) = 0, \quad \text{for} \quad \Gamma = \langle \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \rangle \subset SL_2(\mathbf{Z}).$$

To compute the cohomology groups in (6), we extend the framework of Section 1 relating group cohomology and automorphic representations: the main difference is that the groups

$$H^*(\Gamma, \mathbf{C}) \quad \text{are replaced by} \quad H^*(\Gamma, V)$$

where the coefficient system is no longer trivial, or finite dimensional, but is an infinite-dimensional topological vector space V . For these representations and when the ambient group G is a cover of SL_2 , we extend a result of Bunke-Olbrich [BO98] and prove a Matsushima's formula (1) for infinite-dimensional coefficients.

Theorem 9 ([GGV]). *Let Γ be a lattice in a covering group of $SL_2(\mathbf{R})$, and let W be an infinite-dimensional irreducible smooth Fréchet representation of G with infinitesimal character λ . Then there is a unique complementary representation W^{cl} such that*

$$H^0(\Gamma, W^*) = m(\Gamma, W), \quad H^1(\Gamma, W^*) = m_{\text{cusp}}(\Gamma, W^{cl}),$$

where $m_{\text{cusp}}(\Gamma, V)$ denotes the multiplicity of V in the cuspidal spectrum at level Γ .

We give two proofs of this: one based on [BO98] using resolutions, and one, in the spirit of Matsushima, using $(\mathfrak{g}, K)$ cohomology. The crux in both cases is the surjectivity of a differential operator on the space of automorphic forms, which we prove using the spectral decomposition and a truncation argument.

Specialization of this theorem to $W = \mathcal{S}$ gives the desired dimensions from (6) and allows us to deduce Theorem 8. Moreover, changing the coefficients to other spaces of functions gives new interpolation theorems:

Theorem 10 ([GGV]). *Let $\mathbf{P}^1(\mathbf{R})$ be the real projective line, with homogenous coordinates $[x : y]$. Then any $\Phi \in C^\infty(\mathbf{P}^1(\mathbf{R}))$ is entirely determined the following family of integrals:*

$$a_n(\Phi) = \int_{\mathbf{R}} \Phi(x, 1) e^{\pi i n x} dx, \quad b_n(\Phi) = \int_{\mathbf{R}} \Phi(1, y) e^{\pi i n y} dy.$$

2.2. Future Work. This work has natural extensions in several directions: we would like to prove the technical Theorem 9 for more degrees of cohomology, and extend the results to a more general class of groups. Additionally, it would be interesting to systematize the relation between cohomology vanishing and interpolation theorems.

For higher-rank groups, work on cohomology with coefficients in infinite-dimensional representations already exists: it is the crux of Franke [FS98]’s result to generalize Matsushima’s formula and relate automorphic representations to cohomology of non-compact Shimura varieties. In this direction, establishing new vanishing, or non-vanishing theorems for pieces of the filtration of the mixed Hodge structure associated to the compactification of the Shimura varieties would be of interest. This was recently accomplished in the case of $\Gamma = Sp_{2g}(\mathbf{Z})$ and the top piece of the weight filtration by [BBC⁺24] using algebro-geometric tools. An interesting project would be an automorphic reinterpretation of their result.

3. L -FUNCTIONS AND GAMMA FACTORS: COLLABORATIVE AND MENTORED PROJECTS

To conclude, I discuss two rather different projects [BGGG⁺23] and [FGGHT23] on invariants of automorphic representations, namely gamma factors and L -functions. Both of these projects were part of organized research workshops: Rethinking Number Theory and Women in Numbers, and gave me the opportunity to broaden my scope mathematically and receive important mentorship. I conclude the section by discussing a new Rethinking Number Theory group, for which I am now the mentor.

3.1. Gamma factors and a converse theorem mod ℓ . We consider the problem of classifying irreducible representations of $GL_n(\mathbf{F}_q)$. We study *gamma factors* $\gamma(\rho, \pi)$, invariants introduced by Piatetski-Shapiro for complex representations ρ of $GL_2(\mathbf{F}_q)$, with π a character of $GL_1(\mathbf{F}_q)$. The construction of $\gamma(\rho, \pi)$ is harmonic-analytic, relies on the realization of ρ as a space of functions, its *Whittaker Model*, and satisfies:

Theorem 11 ([PS83]). *A $\mathbf{C}$ -valued cuspidal representation of $GL_2(\mathbf{F}_q)$ is determined by its gamma factors.*

This statement is a *converse theorem*: such results are motivated by local Langlands correspondences, which are characterized by equalities of gamma factors and L -functions. Macdonald [Mac80] proved such a correspondence for representations of $GL_n(\mathbf{F}_q)$, associating them to equivalence classes of Weil group representations. Moreover, representations of $GL_n(\mathbf{F}_q)$ give rise to depth-zero representations of the corresponding p -adic GL_n , and [Mac80] is compatible with the usual local Langlands in depth zero.

3.1.1. Results. In [BGGG⁺23], we consider coefficients in any characteristic $\ell \nmid q$; in particular, ℓ may divide $|GL_n(\mathbf{F}_q)|$. In this situation, Vigneras [Vig88] has extended many results from the complex case. Using these, we first construct mod ℓ gamma factors. Then, exploiting their explicit description as Gauss sums in the case of $GL_2(\mathbf{F}_q)$, we compute counterexamples to the converse theorem on SAGE.

Example 12. When $(\ell, q) = (2, 5), (3, 7), (11, 23),$ or $(29, 59)$, there are non-isomorphic mod ℓ cuspidal representations of $GL_2(\mathbf{F}_q)$ whose gamma factors agree.

This suggests that for mod ℓ representations, one needs to modify the naïve construction. We do this by replacing the Whittaker model of ρ valued in the constants $\bar{\mathbf{F}}_\ell$ by its analogue consisting of functions valued in an Artinian algebra $R(\pi)$ of endomorphisms associated to π . With this new gamma factor, we find:

Theorem 13 ([BGGG⁺23]). *Let ρ be a mod ℓ cuspidal representation of $GL_n(\mathbf{F}_q)$. Then ρ is entirely determined by its $R(\pi)$ -valued γ factors $\gamma(\rho, \pi)$, as π runs over cuspidal representations of $GL_{n-1}(\mathbf{F}_q)$.*

The crux of the proof is establishing a completeness of $R(\pi)$ -valued Whittaker models, which generalizes linear independence of characters: we do this by analyzing ρ and π via their Bernstein-Zelevinsky derivatives.

3.2. Signs in Fourier coefficients of modular forms and zeros of the L-function. This project concerns classical modular Hecke eigenforms f of weight k , given by a Fourier expansion

$$f(z) = \sum_{n=1}^{\infty} a_n(f) n^{\frac{k-1}{2}} e^{\pi i n z},$$

with z in the complex upper half-plane. The Fourier coefficients $a_n(f)$ are multiplicative and the Ramanujan conjecture for SL_2 , proved by Deligne [Del74] for modular forms of weight $k \geq 2$, states that

$$(7) \quad |a_n(f)| \leq \tau(n),$$

for $\tau(n)$ the number of positive divisors of n . We study cancellation between the $a_n(f)$, i.e. of the growth of

$$S(f, x) := \sum_{n \leq x} a_n(f).$$

The best bounds independent of f were proved by Lamzouri [Lam19]: under the GRH for the L -function $L(s, f)$, he showed that $S(f, x) = o(x \log x)$ as $\frac{\log x}{\log \log k} \rightarrow \infty$.

3.2.1. Results. The work [FGGHT23] with Claire Frechette, Alia Hamieh, and Naomi Tanabe, shows that the sums $S(f, x)$ are $o(x \log x)$ for $x \geq k^\epsilon$, under assumptions both weaker and more explicit than GRH:

Theorem 14 ([FGGHT23]). *Let $f \in S_k(N)$ be a primitive cusp form. Let ϵ and T be real numbers with $\epsilon \geq (\log k)^{-1/8}$ and $1 \leq T \leq (\log k)^{1/200}$. Suppose that the region*

$$R(T) = \left\{ s : \Re(s) \geq \frac{3}{4}, |\Im(s)| \leq T + \frac{1}{4} \right\}$$

contains no more than $\epsilon^2 \log k / 5000$ zeroes of $L(s, f)$. Then for all $x \geq k^\epsilon$, we have $|S(f, x)| \ll (x \log x) / T$.

Our proof relies on recent developments in analytic number theory pioneered by Granville–Soundararajan, whose analogous result [GS18] on character sums was the inspiration for the project. The framework, known as *pretentious number theory*, applies naturally to bounded functions. Recently the key input, Halazs’ Theorem, was extended by Mangerel [Man23] to multiplicative functions satisfying (7).

The strategy is to compare $a_f(n)$ and n^{it} for $t \in \mathbf{R}$: the latter see very little cancellation, and for good t the twisted sums $S(f, x, t) = \sum_{n \leq x} a_f(n) n^{it}$ remain close enough to $S(f, x)$. This is useful because:

$S(f, x, t)$ and t -translates of the L -function $L(s, f)$ are related by the Fourier transform.

This allows us to use Plancherel’s formula to compare $S(f, x)$ and a suitable shift $L(s + it, f)$. By assuming for contradiction that $S(f, x)$ is large, we first produce a lower bound involving $L(s + it, f)$, which we express in terms of its zeros using Hadamard’s formula to obtain our result.

3.3. Future Work: Vanishing of Cup Products in Cohomology. Though not central to my research agenda, the two projects described above were very beneficial for my development as a researcher, including for developing expertise in tools, such as the Bernstein-Zelevinsky classification, which played a central role in my main projects. Eager to continue this tradition for other young researchers, I became a mentor for RNT6 in June 2025.

The project, joint with Esme Rosen, Ajmain Yamin, and Zecheng Yang, is to develop an explicit computational framework for the cup product in $(\mathfrak{g}, K)$ -cohomology. We have implemented the computation of the Betti Numbers of cohomological representations of classical groups in Sage, based on Vogan-Zuckerman’s classification [VZ84] and the identification between the cohomology of the trivial representation and that of the compact dual of the symmetric space. We have now moved on to giving explicit descriptions of the cup product of two cohomological $(\mathfrak{g}, K)$ -modules, with the aim of characterizing archimedean obstructions forcing the vanishing of the cup product, as suggested in [Clo92].

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